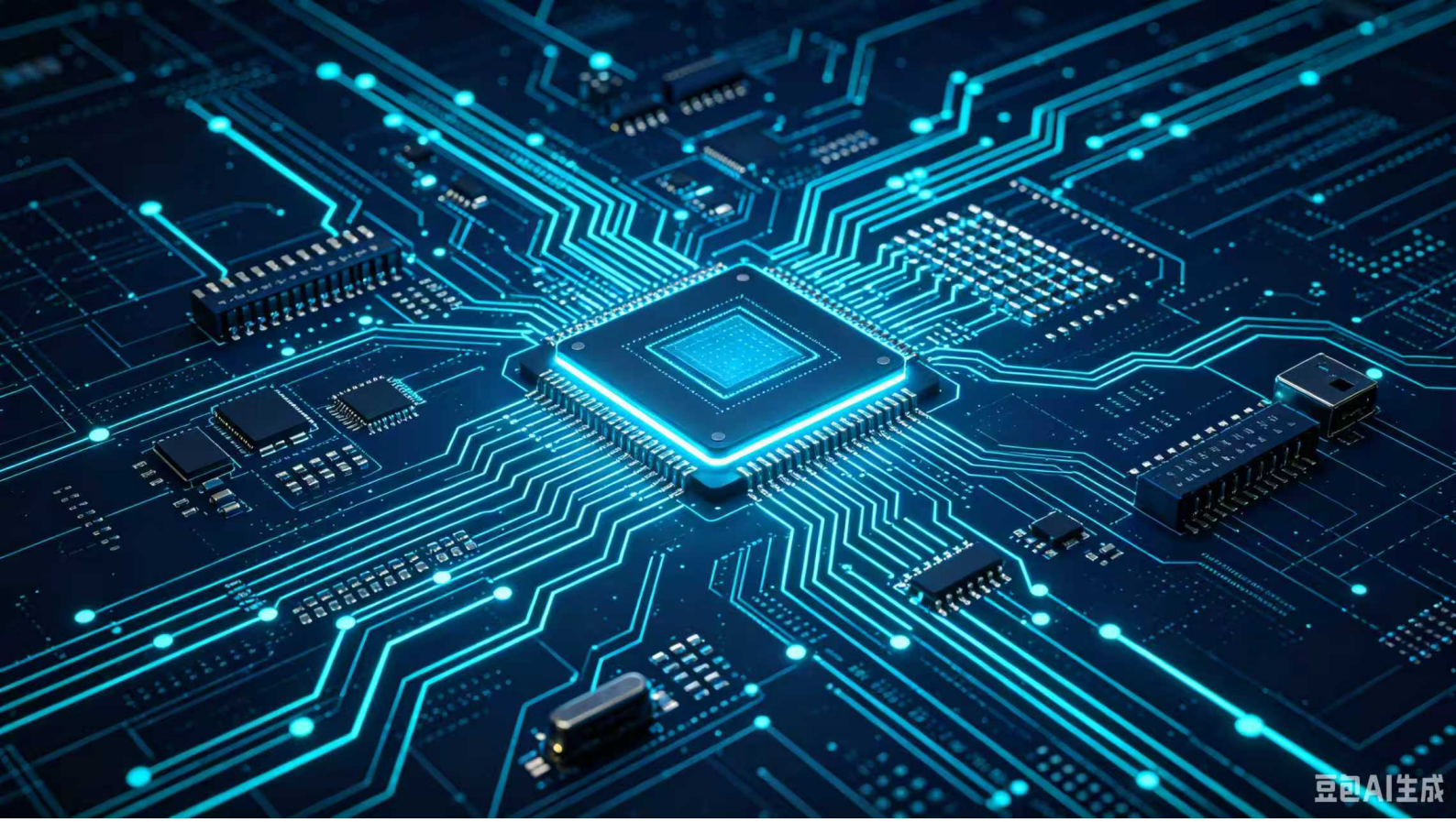


12. Inductors and Inductance

Science and everyday life cannot and should not be separated.
– *Rosalind Franklin, letter to her father, Ellis Franklin; the opening "But" is omitted*

12.1 Inductance

Previously we have introduced a device that can store electric field energy, i.e. the capacitor. There is also a device that can store magnetic field energy as a counterpart, namely an inductor.

Capacitor–Inductor duality. Just as a capacitor stores energy in its electric field, an inductor stores energy in its magnetic field. The basic capacitor is a parallel-plate arrangement producing a uniform electric field; the basic inductor is a long solenoid producing a uniform magnetic field (ignoring fringing). Their defining relations are dual (all derived below):

$$C = \frac{Q}{V}, \qquad L = \frac{N\Phi_B}{i},$$
$$U_E = \frac{Q^2}{2C}, \qquad U_B = \frac{1}{2}Li^2,$$
$$u_E = \frac{1}{2}\epsilon_0 E^2, \qquad u_B = \frac{B^2}{2\mu_0}.$$

where u_E is the electric-field energy density and u_B is the magnetic-field energy density. The SI unit of inductance is the tesla–square meter per ampere ($\mathrm{T{\cdot}m^2/A}$), called the **henry** (H), after the American physicist Joseph Henry, who independently discovered electromagnetic induction around the same time as Faraday:

$$1\ \mathrm{H} = 1\ \mathrm{T} \cdot \mathrm{m}^2/\mathrm{A}.$$

If we look at a solenoid, the magnetic field inside can be easily obtained by applying Ampère's law (ignoring the fringing):

$$B = \mu_0 n i$$

where n is the number of turns per unit length of the solenoid, and i is the current.

Therefore, the magnetic flux through one turn (that is, through the cross section of area S) is:

$$\Phi_B = \mu_0 n S i$$

The unit of magnetic flux is the weber, $1\ \mathrm{Wb} = 1\ \mathrm{T\,m^2}$. The quantity $\mu_0 n S = \Phi_B/i$ depends on the geometry of the solenoid only; therefore Φ_B is proportional to i. The solenoid is a typical example of an **inductor**, and we define the **inductance** as $L = N\Phi_B/i$, where N is the total number of turns. Therefore the inductance L is a measure of the total magnetic flux ($N\Phi_B$) produced by the inductor per unit of current in the solenoid. Thus, if the length of the solenoid is l, we have:

$$L = \frac{N\Phi_B}{i} = \frac{nl\mu_0 n S i}{i} = \mu_0 n^2 l S$$

If the solenoid is filled with a magnetic material of relative permeability μ_r (Chapter 10chapter.10), the field inside becomes $B = \mu_r \mu_0 n i$ and the inductance is multiplied by μ_r; this is why inductors are often wound on iron or ferrite cores.

12.2 Self-Induction

A current i in an inductor L produces a total magnetic flux $N\Phi_B$ in the coil. According to Faraday's law of induction, if the current i varies in time, an induced emf $\mathcal{E}_L$ would appear in the inductor:

$$\mathcal{E}_L = -N\frac{d\Phi_B}{dt} = -L\frac{di}{dt}$$

According to Lenz's law, the direction of the induced current must oppose the change of the flux in the inductor. Thus, the induced emf $\mathcal{E}_L$ should oppose the change in current. Such a process is called *self-induction*.

In our daily life, self-induction is not so difficult to find. If you pull the plug of an appliance with a motor or a transformer while it is running, you may see a small spark at the socket. The current in the windings cannot stop at once: as the contacts separate, the self-induced emf rises until the current jumps the gap. The shock that the prongs of a cheap charger can give just after unplugging can have a different cause, such as charge left on a capacitor across its input.

Potential across an inductor. When a self-induced emf appears in an inductor, we cannot define an electric potential *within* the inductor itself, where the flux is changing. However, potentials can be defined at the terminals (assumed outside the changing-flux region). For an ideal inductor (negligible wire resistance), the magnitude of the terminal potential difference V_L equals the magnitude of the self-induced emf $\mathcal{E}_L$:

$$|V_L| = L\left|\frac{di}{dt}\right|.$$

With the current i counted positive from terminal a to terminal b through the inductor, $V_a - V_b = L\,di/dt$: while the current grows, the potential falls along it; while it decays, the potential rises. This is the term $-L\,di/dt$ in the loop rule of the next section. It follows from Faraday's law round the loop made by the winding and a path between its terminals that stays outside the changing flux and does not go round the coil.

12.3 RL Circuits

We now consider a circuit consisting of a resistor R and an inductor L connected in series. In addition, an emf $\mathcal{E}$ is also connected.

Applying the loop rule, we have:

$$\mathcal{E} - iR - L\frac{di}{dt} = 0$$

or

$$\frac{\mathcal{E}}{R} = \frac{L}{R}\frac{di}{dt} + i$$

We define $\tau_L = \frac{L}{R}$, the **inductive time constant**. For a circuit closed at $t = 0$ with $i(0) = 0$, the solution to the above equation is:

$$i = \frac{\mathcal{E}}{R}\left(1 - e^{-t/\tau_L}\right)$$

If the source is then removed and the loop is closed through R alone, the current decays from its value $i_0 = \mathcal{E}/R$ at that instant as:

$$i = i_0 e^{-t/\tau_L}$$

During the process, the work done by the emf is partially dissipated by the resistor, and the rest stored in the inductor. Once the steady state is reached, the current stops varying, and from then on all the work done by the emf is dissipated in the resistor. Rewriting the loop rule as:

$$i\mathcal{E} = i^2 R + iL\frac{di}{dt}$$

Therefore, the rate at which energy is stored in the inductor is:

$$\frac{dU_B}{dt} = Li\frac{di}{dt}$$

So the total energy stored in the inductor can be written as:

$$U_B = \int_0^t Li'\frac{di'}{dt'}dt' = \int_0^i Li'\,di' = \frac{1}{2}Li^2$$

It is similar to the expression for the stored energy in a capacitor $U_E = \frac{q^2}{2C}$; the energy stored in the magnetic field is taken up again in Section 12.4section.12.4.

Comparison of RL and RC circuits:

	RC circuit	RL circuit
Time constant	$\tau_C = RC$	$\tau_L = L/R$
Charging/rise	$q = Q(1 - e^{-t/\tau_C})$	$i = \frac{\mathcal{E}}{R}(1 - e^{-t/\tau_L})$
Energy stored	$U_E = \frac{Q^2}{2C}$	$U_B = \frac{1}{2}Li^2$

where $Q = C\mathcal{E}$ is the final charge.

12.4 Energy Stored in a Magnetic Field

The stored energy in a capacitor can be viewed as the total energy stored in the electric field. The total energy stored in an inductor can also be regarded as energy stored in the magnetic field. We now take the ideal solenoid of length l (Section 12.1section.12.1) and n turns per unit length. If its cross sectional area is S, we have derived:

$$L = \mu_0 n^2 l S$$

Therefore, the total energy stored is:

$$U_B = \frac{1}{2}Li^2 = \frac{1}{2}\mu_0 n^2 l S i^2 = \frac{1}{2\mu_0}(\mu_0 n i)^2 (Sl)$$

Noticing that the magnetic field of the solenoid is $B = \mu_0 n i$, and Sl is the volume occupied by the magnetic field (again, ignoring the fringing), we can define:

$$u_B = \frac{1}{2\mu_0}B^2$$

Although we obtained u_B for the uniform field of a solenoid, it holds at every point of any magnetic field in vacuum, as u_E does for any electric field (Chapter 7chapter.7). The energy stored in a magnetic field can therefore be found either from the inductance and the current, or by integrating u_B over the region the field occupies. For the solenoid the two are the same calculation read in opposite directions; for the coaxial cable and the toroid of Exercises 1 and 2, where the field varies from point to point, they are independent, and they agree.

12.5 Mutual Induction

We now consider two coils L_1 and L_2 near each other; Figure 12.1figure.12.1 shows the general arrangement, drawn as a schematic with the coils apart along their axis. A current i_1 in L_1 produces a magnetic field, part of which enters L_2 and causes magnetic flux in L_2. We define the **mutual inductance** M_{21} by:

$$M_{21} = \frac{N_2 \Phi_{21}}{i_1}$$

M_{21} is the flux linkage in L_2 per unit current in L_1. Thus, if i_1 varies, there will be an emf produced in L_2:

$$\mathcal{E}_{21} = -N_2 \frac{d\Phi_{21}}{dt} = -M_{21}\frac{di_1}{dt}$$

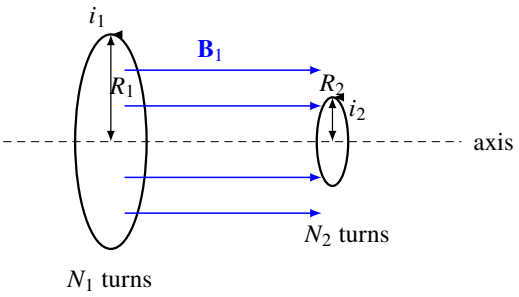


Figure 12.1: Two parallel coaxial coils, drawn as a schematic. A current i_1 in the larger coil (radius R_1, N_1 turns) produces the magnetic field $\mathbf{B}_1$ that threads the smaller coil (radius R_2, N_2 turns), which carries a current i_2. The flux linkage $N_2\Phi_{21}$ produced in the smaller coil by i_1 defines the mutual inductance $M_{21} = N_2\Phi_{21}/i_1$.

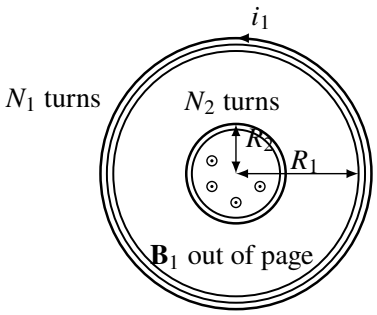


Figure 12.2: Two coaxial parallel circular coils with N_1 and N_2 turns and radii R_1 and R_2, drawn as seen along their common axis. The smaller coil is centered on the axis of the larger one and lies in the same plane, so that the magnetic field of the larger coil at its center, $B_1 = \mu_0 N_1 i_1/(2R_1)$, points along the axis and is uniform over the smaller coil, giving a mutual inductance $M_{21} = \mu_0 N_1 N_2 \pi R_2^2/(2R_1)$.

Example 12.1: Mutual inductance between two parallel coils

Two circular coils share an axis and lie in the same plane: the larger has radius R_1 and N_1 turns, and the smaller, at its center, has radius $R_2 \ll R_1$ and N_2 turns (Figure 12.2figure.12.2). We calculate the total flux in the smaller coil due to the larger coil, assuming that the smaller coil is sufficiently small and the magnetic field through it can be regarded as uniform. The magnetic field at the center of the larger coil is given by:

$$B_1 = \frac{\mu_0}{2}\frac{N_1 i_1}{R_1}$$

This can be immediately derived from our previous calculation of the magnetic field along the axis of a single circular loop (Example 9.1example.9.1). Therefore, each loop of the smaller coil contributes:

$$\Phi_{21} = \pi R_2^2 B_1 = \frac{\mu_0}{2}\frac{N_1 i_1 \pi R_2^2}{R_1}$$

to the magnetic flux. This leads to:

$$M_{21} = \frac{N_2 \Phi_{21}}{i_1} = \frac{\mu_0 N_1 N_2 \pi R_2^2}{2R_1}$$

As the example above shows, the mutual inductance appears to be a purely geometric quantity. In vacuum it depends only on the shapes and positions of the two circuits; a magnetic core changes it (Chapter 10chapter.10).

Similarly, we can also define M_{12} as the ratio between total flux in L_1 caused by current i_2 in L_2. We can prove $M_{12} = M_{21}$ for two fixed coils. Therefore, we can ignore the subscripts and simply call it M. The mutual inductance can be positive or negative: its sign depends on the positive senses chosen for the two currents, and it changes if either sense is reversed (Chapter 13chapter.13).

Proof that $M_{12} = M_{21}$ always holds.

We shall do this by assuming that we have two coils, L_1 and L_2, connected to two emf devices $\mathcal{E}_1$ and $\mathcal{E}_2$, respectively. The emf device is specially designed such that it can hold the current through it at any prescribed value. The two coils are close enough so that there is mutual inductance in between. The two coils are fixed, and their resistance is negligible. We consider the work done by the emf devices in the process that we bring the current through L_1 from 0 to i_1 and the current through L_2 from 0 to i_2. In particular, we consider these two paths:

Path 1: We bring L_1 from 0 to i_1 first, keeping the current through L_2 to be 0; then we keep L_1 at i_1, bring L_2 from 0 to i_2.

In the first half, the work done by the first emf is:

$$W_1 = \frac{1}{2} L_1 i_1^2$$

and since the current through L_2 is kept at 0, the work done by the second emf is simply 0.

In the second half, the work done by the second emf is:

$$W_2 = \frac{1}{2} L_2 i_2^2$$

However, when the current is changing in the second coil, an emf is also induced in the first coil due to the mutual inductance: $\mathcal{E}_1' = -M_{12}\frac{di_2}{dt}$. In order to maintain the i_1 current, the first emf must compensate this change with power

$$P = i_1 \mathcal{E}_1 = M_{12} i_1 \frac{di_2}{dt}$$

This means that, by the time L_2 reaches i_2, the additional work done by the first emf is:

$$W_1' = M_{12} i_1 i_2$$

Thus the total work done is

$$W = W_1 + W_2 + W_1' = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M_{12} i_1 i_2$$

Path 2: We now bring L_2 from 0 to i_2 first, keeping the current through L_1 to be 0; then we keep L_2 at i_2, bring L_1 from 0 to i_1.

It is not difficult to repeat the above analysis in this situation and obtain that the total work done is now:

$$W' = W_1 + W_2 + W_2' = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M_{21} i_1 i_2$$

We note that the work done by the two processes leads to the same final state, and there is no dissipation process due to resistance involved. Therefore all the work done is stored as magnetic energy, $\int B^2/2\mu_0 \, dV$. That energy depends only on the final field, and the final field is fixed by the final currents i_1 and i_2, since the fields of the two coils add. The two works are therefore equal, and $M_{12} = M_{21}$.

Inductance matrix. For a system of two coupled circuits, the flux linkages and induced emfs can be expressed in matrix form:

$$\begin{pmatrix} N_1\Phi_1 \\ N_2\Phi_2 \end{pmatrix} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}, \qquad \begin{pmatrix} \mathcal{E}_{\text{ind},1} \\ \mathcal{E}_{\text{ind},2} \end{pmatrix} = -\begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \frac{d}{dt} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}.$$

The symmetry of the inductance matrix reflects the reciprocity $M_{12} = M_{21}$.

12.6 Exercises

1. Consider two coaxial cylindrical conducting shells of radii R_1 and R_2 ($R_1 < R_2$) and length l. A current I flows down the inner shell and back along the outer one, spread uniformly round each shell. Assume the shells are long enough that boundary effects can be ignored. Calculate the self-inductance of the system.

2. A toroidal coil with an air core has a rectangular cross-section. The inner radius is a, the outer radius is b, and the height is h. The toroid has N turns of wire uniformly wound around it. Find the self-inductance L of the toroid.

3. An infinitely long straight wire carrying current I lies in the same plane as a rectangular loop of dimensions $a \times b$. The near edge of the loop is at distance c from the wire, and the far edge is at distance $b + c$ from the wire.

   (a) Find M_{21}, the flux through the loop per unit current in the wire.

   (b) If $a = 10$ cm, $b = 5$ cm, $c = 3$ cm, calculate M.

   (c) What is the induced emf in the loop if $\mathrm{d}I/\mathrm{d}t = 100$ A/s? Give its magnitude and the sense of the induced current on the side nearer the wire.

4. Two circular loops of radii R_1 and R_2 lie in parallel planes separated by distance d. Their centers are on the same axis perpendicular to both planes. Count both currents positive in the same sense about the axis.

   (a) Show that the mutual inductance of any two loops can be written in the Neumann form

   $$M = \frac{\mu_0}{4\pi} \oint \oint \frac{\mathrm{d}\boldsymbol{l}_1 \cdot \mathrm{d}\boldsymbol{l}_2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}.$$

   Hint: write the flux through loop 2 as $\oint \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{l}_2$, with $\boldsymbol{A} = \frac{\mu_0 i_1}{4\pi} \oint \frac{\mathrm{d}\boldsymbol{l}_1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}$. First show that $\nabla \times \boldsymbol{A}$, with this $\boldsymbol{A}$, equals the magnetic field of loop 1 given by the Biot–Savart law; then use Stokes' theorem, $\int_S (\nabla \times \boldsymbol{A}) \cdot \mathrm{d}\mathbf{S} = \oint_C \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{l}$.

   (b) Show that for the case of $d \gg R_1, R_2$, the mutual inductance is

   $$M \approx \frac{\pi \mu_0}{2d^3} R_1^2 R_2^2.$$

   Hint: this part does not follow *immediately* from the Neumann form; use the on-axis field of a single circular loop (Example 9.1example.9.1) at distance d.

5. A long, closely wound air-core solenoid of length ℓ, cross-sectional area S and n turns per unit length has its axis along z and negligible resistance; neglect end effects. Its terminals are A and B, and its current is counted positive when it enters at A, leaves at B, and circulates counterclockwise as seen from $+z$. An ideal current source makes the current $i(t) = I_0(1 - 2t/T)$ for $0 \le t \le T$, where $I_0 > 0$ and $T > 0$, so that the current reverses from $+I_0$ to $-I_0$. Let L be the solenoid's self-inductance.

   (a) Find the self-induced emf $\mathcal{E}_L$, counted positive along the winding in the positive sense, and the potential difference $V_A - V_B$. Which terminal is at the higher potential, and does this change when the current passes through zero?

   (b) Find the magnetic energy stored in the solenoid at $t = 0, T/2$ and T, and the work done by the source on the solenoid during $0 \le t \le T/2$ and during $T/2 \le t \le T$.

6. A coil of inductance L and resistance r (treat it as an ideal inductor L in series with a resistor r) has terminals A and B. A resistor R is connected directly between A and B. An ideal battery of emf $\mathcal{E}$ is also connected between A and B, through a switch, with its positive terminal on the side of A. The switch has been closed for a long time and is opened at $t = 0$; after that the coil and R form a closed loop, and the battery is disconnected. The quantities L, r, R and $\mathcal{E}$ are all positive.

   (a) Just before $t = 0$, find the current through the coil, the current through R, and $V_A - V_B$.

   (b) For $t > 0$, find the current through the coil, and find $V_A - V_B$ just after $t = 0$. For which values of R is $|V_A - V_B|$ just after $t = 0$ larger than $\mathcal{E}$?

   (c) Find the heat produced in R after $t = 0$, and show that the total heat produced in R and in the coil equals the magnetic energy stored in the coil at $t = 0$.

7. A long coaxial cable, with its axis along z, consists of a solid inner conductor of radius a, which carries a current $I > 0$ along $+z$, spread uniformly over its cross section, and a thin coaxial cylindrical shell of radius $b > a$, which carries the same current back, spread uniformly round it. Both conductors are nonmagnetic, so the energy density $B^2/2\mu_0$ holds at every point, inside the metal as well as between the conductors. Consider a length l of the cable and neglect end effects.

   (a) Find the magnitude and direction of the magnetic field at a distance r from the axis, for $r < a$, $a < r < b$ and $r > b$, and the magnetic energy stored in the length l inside the inner conductor and between the conductors.

   (b) Using $U_B = \frac{1}{2}LI^2$, find the self-inductance L of the length l.

   (c) A student instead computes the magnetic flux through a flat rectangle of length l that lies in a plane containing the axis and extends from the axis to $r = b$, and divides it by I. Find the result, and explain why it is not the self-inductance.

8. A thin toroidal iron core has mean circumference ℓ, measured along its center line, and cross-sectional area S, with $\sqrt{S} \ll \ell$. A narrow air gap of width $g \ll \sqrt{S}$ is cut straight across it. N turns of wire, wound uniformly around the core, carry a current i. Treat the iron as linear, with relative permeability μ_r ($\mathbf{B} = \mu_r\mu_0\mathbf{H}$, Chapter 10), and assume that all the flux follows the core and crosses the gap straight, with no leakage and no fringing, so that $\mathbf{B}$ has the same magnitude over the whole cross section.

   (a) Explain why the magnitude B of the field is the same in the iron and in the gap, and find B.

   (b) Find the self-inductance L of the winding.

   (c) Find the fraction of the stored energy $\frac{1}{2}Li^2$ that is in the gap. Evaluate it for $\mu_r = 5000$, $\ell = 0.20$ m and $g = 1.0$ mm.

9. Two fixed, long, thin-wound solenoids share the z axis and have the same length ℓ. The inner one has radius a and n_1 turns per unit length; the outer one has radius $b > a$ and n_2 turns per unit length. Neglect end effects and the resistance of the windings, and take the space inside and around them to be vacuum. A current i_1 or i_2 is counted positive when it produces a field along $+\hat{z}$. Ideal sources can change either current slowly while holding the other fixed.

   (a) Find the self-inductances L_1 and L_2 and the mutual inductances M_{12} and M_{21}.

   (b) The currents are brought from zero to final values $I_1 > 0$ and $I_2 > 0$ in two ways: first i_1 to I_1 and then i_2 to I_2, or first i_2 to I_2 and then i_1 to I_1. Find the total work done

by the two sources in each case, including the work of the source that holds its current fixed while the other changes.

(c) Find the final magnetic energy by integrating $B^2/2\mu_0$ over the regions $r < a$ and $a < r < b$, and compare it with your answer to (b).